Entanglement and Special Relativity: The Standard View

I have put the words “forces” and “compels” in quotes because while they convey the sentiment our classical intuition longs for, their precise meaning in this context is critical to whether or not we are in for even more of an upheaval. With their everyday definitions, these words conjure up an image of volitional causality: we choose to do something here so as to *cause* or *force* a particular something to happen over there. If that were the right description of how the two photons are interrelated, *special relativity would be on the ropes.* The experiments show that from the viewpoint of an experimenter in the laboratory, at the precise moment one photon’s spin is measured, the other photon immediately takes on the same spin property. If something were traveling from the left photon to the right photon, alerting the right photon that the left photon’s spin had been determined through a measurement, it would have to travel between the photons instantaneously, conflicting with the speed limit set by special relativity.

The consensus among physicists is that any such apparent conflict with special relativity is illusory. The intuitive reason is that even though the two photons are spatially separate, their common origin establishes a fundamental link between them. Although they speed away from each other and become spatially separate, their history entwines them; even when distant, they are still part of one physical system. And so, it’s really not that a measurement on one photon forces or compels another distant photon to take on identical properties. Rather, the two photons are so intimately bound up that it is justified to consider them—even though they are spatially separate—as parts of one physical entity. Then we can say that one measurement on this single entity—an entity containing two photons—affects the entity; that is, it affects both photons at once.

While this imagery may make the connection between the photons a little easier to swallow, as stated it’s vague—what does it really mean to say two spatially separate things are one? A more precise argument is the following. When special relativity says that nothing can travel faster than the speed of light, the “nothing” refers to familiar matter or energy. But the case at hand is subtler, because it doesn’t appear that any matter or energy is traveling between the two photons, and so there isn’t anything whose speed we are led to measure. Nevertheless, there is a way to learn